ISPRS Journal of Photogrammetry and Remote Sensing 218 (2024) 87–101

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

Improving crop type mapping by integrating LSTM with temporal random masking and pixel-set spatial information

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ARTICLE INFO

Keywords:
Crop type mapping
Temporal random masking
Spatial information aggregation
Early season crop classification
Satellite image time series

ABSTRACT

Accurate and timely crop type classification is essential for effective agricultural monitoring, cropland management, and yield estimation. Unfortunately, the complicated temporal patterns of different crops, combined with gaps and noise in satellite observations caused by clouds and rain, restrict crop classification accuracy, particularly during early seasons with limited temporal information. Although deep learning-based methods have exhibited great potential for improving crop type mapping, insufficient and noisy training data may lead them to overlook more generalizable features and derive inferior classification performance. To address these challenges, we developed a Mask Pixel-set SpatioTemporal Integration Network (Mask-PSTIN), which integrates a temporal random masking technique and a novel PSTIN model. Temporal random masking augments the training data by selectively removing certain temporal information to improve data variability, enforcing the model to learn more generalized features. The PSTIN, comprising a pixel-set aggregation encoder (PSAE) and long short-term memory (LSTM) module, effectively captures comprehensive spatiotemporal features from time-series satellite images. The effectiveness of Mask-PSTIN was evaluated across three regions with different landscapes and cropping systems. Results demonstrated that the addition of PSAE in PSTIN significantly improved crop classification accuracy compared to a basic LSTM, with average overall accuracy (OA) increasing from 80.9% to 83.9%, and the mean F1-Score (mF1) rising from 0.781 to 0.818. Incorporating temporal random masking in training led to further improvements, increasing average OA and mF1 to 87.4% and 0.865, respectively. The Mask-PSTIN significantly outperformed traditional machine learning and deep learning methods (i.e., RF, SVM, Transformer, and CNN-LSTM) in crop type mapping across all three regions. Furthermore, Mask-PSTIN enabled earlier and more accurate crop type identification before or during their developing stages compared to machine learning models. Feature importance analysis based on the gradient backpropagation algorithm revealed that Mask-PSTIN effectively leveraged multi-temporal features, exhibiting broader attention across various time steps and capturing critical crop phenological characteristics. These results suggest that Mask-PSTIN is a promising approach for improving both post-harvest and early-season crop type classification, with potential applications in agricultural management and monitoring.

1. Introduction

The world's population has grown triple larger than in the mid-twentieth century (Lobell et al., 2013), rapidly increasing global food demands. Current agricultural practices to meet these rising food needs, such as expanding croplands, increasing fertilizer use, and intensifying

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https://doi.org/10.1016/j.isprsjprs.2024.10.013
Received 24 April 2024; Received in revised form 10 October 2024; Accepted 11 October 2024
Available online 19 October 2024
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cropping practices, may threaten the environment (Bennetzen et al., 2016). Sustainable agricultural management aims to balance food production and environmental protection (Tilman et al., 2011; van Dijk et al., 2021). Accurate and timely spatial distribution information on crop types is fundamental data for yield estimation, cropping patterns management, and pest control (Cai et al., 2018; van Loon et al., 2019), all of which are vital for sustainable agricultural practices (Asam et al., 2022; Xin et al., 2020). However, obtaining crop information through traditional field surveys and census data is time-consuming and expensive, especially for large-scale and historical monitoring (Becker-Reshef et al., 2010; Zhong et al., 2014).

The rapidly developing remote sensing techniques have been recognized as a cost-effective alternative for obtaining area and spatial distribution information of crop types (Hu et al., 2021; Lou et al., 2023; Skakun et al., 2017; Xia et al., 2022). Considerable efforts have been devoted to map crop types using satellite data and these approaches can be categorized into three groups: threshold-based methods, machine learning methods, and deep learning methods (Becker-Reshef et al., 2010; Boryan et al., 2011; Jiao et al., 2014; Weiss et al., 2020). Threshold-based methods identify crop types based on pre-determined classification rules and crop-specific phenological characteristics. For instance, Xiao et al. (2005) effectively classified paddy rice during the transplanting phase using the rule that the Land Surface Water Index (LSWI) is greater than the Enhanced Vegetation Index (EVI). However, the performances of these methods were highly dependent on the spatiotemporal stability of crop phenology. Due to the changes in climate and agricultural practices, the phenological characteristics of crop types may vary significantly across space and time, leading to poor generalizability and transferability of pre-determined classification rules across different regions and years. Machine learning methods, such as random forest, decision tree, and support vector machine, have been successfully applied in classifying different crop types over large areas due to their ability to automatically learn relationships between satellite-derived features and crop types (Blickensdörfer et al., 2022; Fowler et al., 2020; Hu et al., 2021; Yoon and Kim, 2020). Nevertheless, these methods typically rely on shallow and hand-crafted features and struggle to capture complex temporal patterns during crop growth, limiting their performance and generalizability for mapping crop types (Zhong et al., 2019).

In contrast, deep learning methods have proven effective in extracting deep representative features from multi-temporal remote sensing data for crop type mapping (Asadi and Shamsoddini, 2024; Cai et al., 2023b, 2024; Tang et al., 2024; Wang et al., 2023). Among these methods, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are widely used approaches according to previous studies (Wang et al., 2022; Zhang et al., 2024). Specifically, CNNs primarily extract texture information from images in the spatial domain (Zhong et al., 2019), while RNNs are specialized in modeling temporal dependencies within sequential data, making them particularly effective in capturing the phenological dynamics of crops from time-series satellite data (Feng et al., 2023; Xu et al., 2023). Long short-term memory (LSTM) units, a variant of RNN, have shown a superior ability to memorize long-term temporal information by introducing a gated mechanism, enabling the successful applications of LSTM-based methods in large-scale crop type mapping (Wang et al., 2020; Zhong et al., 2019). Crisóstomo De Castro Filho et al. (2020) compared the performance of LSTM and bidirectional LSTM (BiLSTM) in rice mapping in Brazil, finding that BiLSTM exhibited better results by learning more comprehensive information from both forward and backward directions. Xu et al. (2020) further improved LSTM's ability to explore the temporal features by incorporating the attention mechanism, significantly outperforming traditional machine learning models in mapping corn and soybean distribution over large areas. Despite these advancements, the practical application of time-series satellite data is often influenced by noise and gaps due to cloud cover and rain, leaving it challenging to explore robust and invariant crop features (Zhao et al., 2021). The challenge is exacerbated by the difficulty in collecting a substantial number of samples, which can cause the model to overfit a specific temporal pattern or noise present in the training data, ultimately reducing classification performance (Salman and Liu, 2019; Zhao, 2017). Moreover, early-season crop mapping, which refers to the classification of crop types using limited satellite observations during the early stages of the growing season, generally shares similar classification methods with post-harvest crop mapping but uses fewer available satellite images (Wang et al., 2024; You and Dong, 2020). Nevertheless, it is particularly challenging compared to post-season classification due to the less temporal information available for analysis (McNairn et al., 2014; Potgieter et al., 2010; Zhang et al., 2021).

To mitigate these challenges, several studies have explored reconstructing cloud-free time-series observations through temporal interpolation and multi-source data integration, aiming to capture critical phenological information about crops and improve classification accuracy (Wang et al., 2024; Zhou et al., 2024). However, recent studies suggest that these interpolation processes may not reliably reconstruct time series, offering limited improvement in crop type mapping and increasing computational costs (Che et al., 2024; Zhao et al., 2021). By contrast, data augmentation techniques have been demonstrated to increase the training data variability and improve models' robustness against noise and gaps. For instance, randomly removing contiguous parts of input images during data augmentation can significantly improve models' performance in handling occlusion issues of image classification (DeVries and Taylor, 2017; Zhong et al., 2020). The combination of different augmentations to process audio sequence data, such as frequency masking and time masking, has been proven to improve speech recognition performance even with limited training data (Alex et al., 2023; Park et al., 2019). As for time-series satellite image analysis, Chen et al. (2022) developed a novel imputation-BiLSTM model that introduced a missing data imputation task, jointly trained with crop classification, to address incomplete time-series satellite data caused by cloud contamination. Garnot et al. (2022) adopted a temporal dropout strategy that randomly removed observations from multi-modal time-series data during training stages to prevent the model from overly relying on a single modality. Inspired by this, Pham et al. (2024a, 2024b) introduced random observation selection to simulate the variations and sparsity of temporal data, thereby improving the models' transferability. Although these augmentation approaches have made significant progress, they were mainly tested in single agricultural landscapes. The generalizability of these methods across diverse crop types and landscapes remains unexplored. Therefore, it is critical to develop an effective data augmentation strategy to address overfitting in limited and low-quality datasets and improve the model's ability to capture crop phenology dynamics.

Furthermore, traditional classification methods were mainly implemented at the pixel level, which not only were easily affected by the "salt-and-pepper" problem but also neglected spatial features during crop growth caused by varying agricultural management practices and planting patterns (e.g., furrows or crop rows) (Jiao et al., 2014; Lin et al., 2022; Valero et al., 2021; Wei et al., 2023). Although the CNNs have been widely used for extracting spatial features from satellite images, their heavy reliance on texture may limit their effectiveness in leveraging spatial information from medium-resolution (10–30 m) satellite images, which may be too coarse to depict the agricultural textural details (Geirhos et al., 2018; Garnot et al., 2019). Recent studies have pointed out that viewing medium-resolution images as unordered sets of pixels and using pixel-set encoders to extract statistical features of spectral distribution within agricultural field parcels can be more effective than traditional CNNs in crop type mapping (Garnot et al., 2020; Weilandt et al., 2023; Yuan et al., 2023). These methods typically depend on agricultural field parcel data as a spatial constraint to ensure that pixels in the pixel-set share the same category information. Despite the increasing availability of open-access agricultural field parcel datasets and the rapid development of field delineation methods (Cai et al.,

2023a; Meyer et al., 2020), obtaining timely and accurate field parcel data remains challenging, particularly in smallholder farming regions like southern China. However, applying pixel-set methods directly to image patches can lead to inferior results due to the potential inclusion of features from different crops within the patch. Therefore, effectively integrating information from spatially neighboring pixels is crucial for capturing spatial information of crops and reducing noise effects in individual pixels, especially in areas characterized by small and heterogeneous landscapes.

In this study, we propose a deep learning method named Mask Pixel-set SpatioTemporal Integration Network (Mask-PSTIN), which leverages the spectral-spatial–temporal information of the time-series satellite images to improve the accuracy of crop type classification. The major contents of our study can be summarized in three aspects: (1) a temporal random masking augmentation strategy, which selectively masks observations at certain time steps from the time-series data, encouraging the model to learn more generalizable and robust features to avoid overfitting; (2) a novel pixel-set aggregation encoder that adaptively aggregates the spectral information from spatially neighbored pixels, integrated with the LSTM units and a temporal attention mechanism to explore the comprehensive spectral-spatial–temporal information from multi-temporal images, improving the accuracy of post-harvest and early-season crop type mapping; (3) based on gradient backpropagation methods, the feature importances were analyzed to interpret the advantages of Mask-PSTIN. Furthermore, we evaluated the effectiveness and generalization of Mask-PSTIN with experiments across three regions characterized by diverse agricultural landscapes and cropping patterns.

2. Methodology

2.1. The overview of the Mask-PSTIN model

The framework of the developed Mask-PSTIN is presented in Fig. 1. Mask-PSTIN consists of a novel classification model (PSTIN) and a data augmentation technique involving temporal random masking. Specifically, PSTIN combines a pixel-set aggregation encoder (PSAE) with a BiLSTM and a temporal attention (AtBiLSTM) module to capture the spatiotemporal patterns of time-series satellite image patches. The random masking technique augments the limited training data by selectively masking observations at certain time steps in the multi-temporal satellite images during the training process, thereby enforcing the model to learn more robust and invariant features of crop types. Detailed information on these two components is provided in Sections 2.2 and 2.3. The code will be publicly available at https://github.com/BruceKai/Mask-PSTIN.

2.2. Temporal random masking

Given the complicated temporal patterns of different crops and the challenges posed by missing data and noise due to poor observations (e.g., clouds or rains), training deep learning models on insufficient and noisy data risks overfitting specific temporal patterns or noise, ultimately resulting in poor performance. To address this issue, we implemented a temporal random masking strategy to augment the training data (Fig. 2). During the training process, for a given time-series satellite

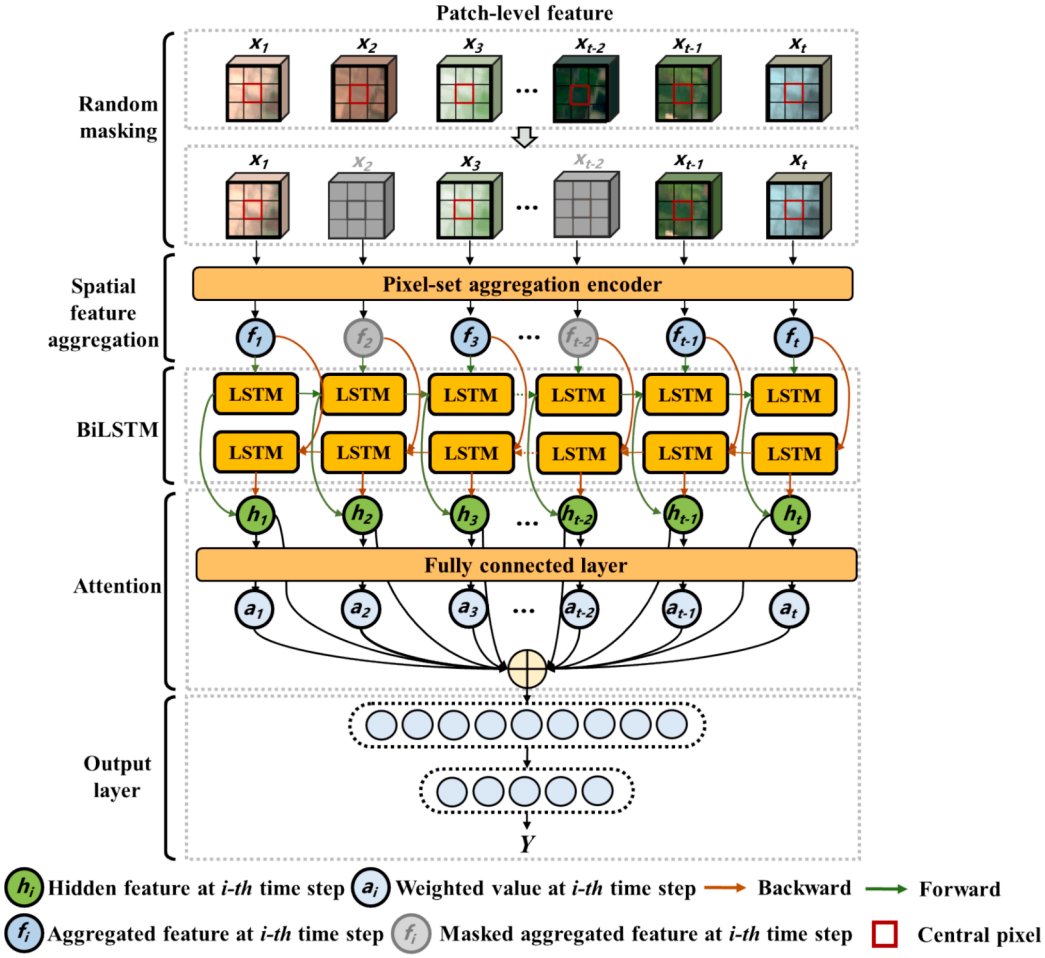


Fig. 1. The overview of Mask-PSTIN for crop type mapping. X_1 to X_t represents the input patch at different time steps. The configuration of the PSTIN is detailed in Part I of Supplementary Material.

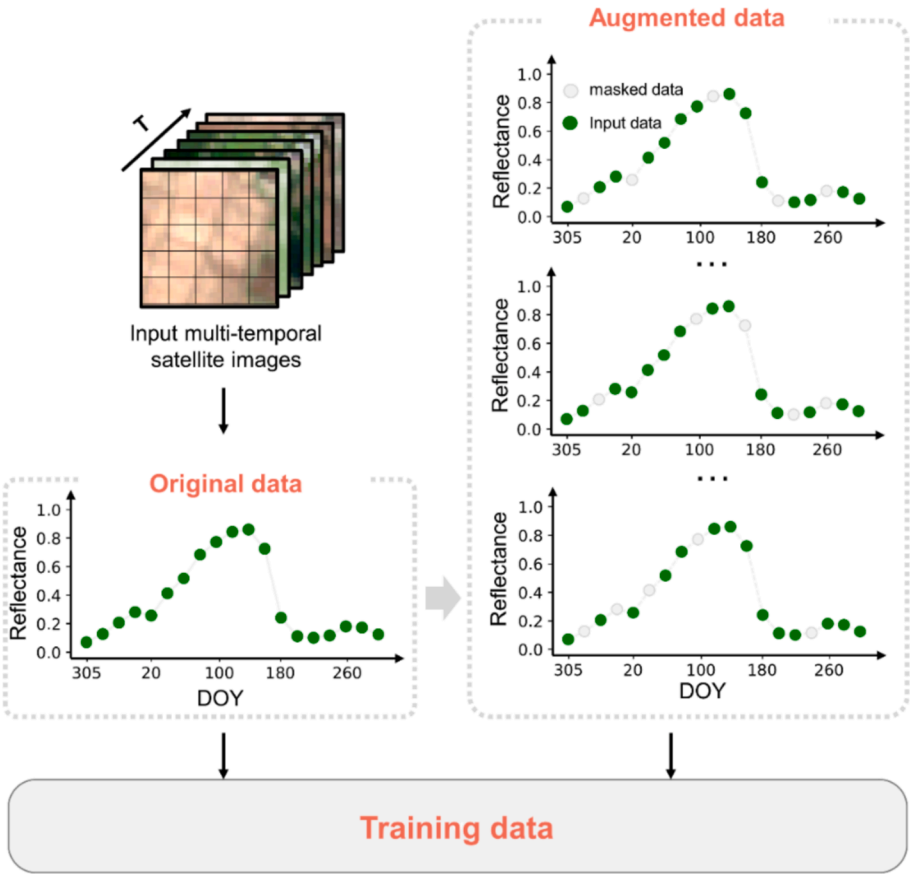


Fig. 2. The example of augmenting training data by temporal random masking technique.

image $[X_1, X_2, \ldots X_T]$ with a sequence length of T, we first generated a random mask probability vector $[M_1, M_2, \ldots M_T]$ of the same length through a random number generator. The masked result at time step t was then derived according to Eq. (1):

$$X_t = \begin{cases} 0, M_t < \tau_m \\ X_t, otherwise \end{cases} \quad (1)$$

where X_t represents the input satellite image at time step t, and M_t is the corresponding randomly generated mask probability. τ_m is a tunable masking threshold parameter ranging from 0 to 1. If the generated probability M_t falls below the threshold τ_m, the corresponding X_t values are padded with zeros. Thus, the τ_m controls the extent of temporal information masked during data augmentation. A higher τ_m value increases the variability of training data, improving the model's robustness against noise or missing data. However, it may also risk underfitting if too much useful temporal information, including key phenological phases, is excessively masked. Conversely, a lower τ_m value preserves more original temporal information, which may limit the model's generalization on intra-class variation. The optimal τ_m varies with specific cropping patterns and phenological characteristics, as detailed in Part III of the Supplementary Material. By randomly masking observations at certain time steps of the input multi-temporal satellite data, we aim to augment the training data and increase its variability. This approach improves the model's capacity to learn more generalized and invariant features, preventing it from overly relying on features at specific time steps for classification and ensuring more robust performance in the face of diverse temporal patterns and noise.

2.3. PSTIN for crop type mapping

To leverage spatial and temporal information of satellite image time series, we developed a novel patch-based classification model PSTIN for identifying different crop types (Fig. 1). The PSTIN processes time-series image patches centered on the target pixel, with a patch size of 5 × 5, as referenced to previous studies (Ienco et al., 2019; Interdonato et al., 2019; Yuan et al., 2022). For each time-series image patch, the PSTIN first uses the PSAE to aggregate the spectral features of all pixels within the patch under the guidance of the central pixel. It then employs the BiLSTM and temporal attention module to explore the temporal dynamics of aggregated spectral-spatial features at different time steps.

2.3.1. Pixel-set aggregation encoder (PSAE)

Previous pixel-set methods for extracting spatial features often directly combine spectral information from all pixels within a patch, which can dilute the representation of the central pixel when surrounding pixels contain different crop information (Garnot et al., 2019). To address this, we developed a novel PSAE module that calculates the similarity between the central pixel and its surrounding pixels, allowing for a weighted aggregation of spectral information from different pixels based on this similarity.

Fig. 3 presents the architecture of the developed PSAE. For an input patch with dimensions of $C_0 \times H \times W$, we first flatten it into a set of $H \times W$ pixels, where each pixel contains C_0 spectral bands. A shared-weight multi-layer perceptron (MLP) with two layers was then employed to extract deeper spectral features for each pixel within the patch. These pixel features were used to generate the key (K) and value (V) vectors, while the central pixel generated a query (Q) vector. By calculating the dot product between the query and key vectors, the PSAE derived similarity scores indicating how closely the surrounding pixels' features aligned with those of the central pixel. These scores were subsequently normalized using a softmax function to generate similarity weights. Finally, the spectral features of surrounding pixels were weighted by

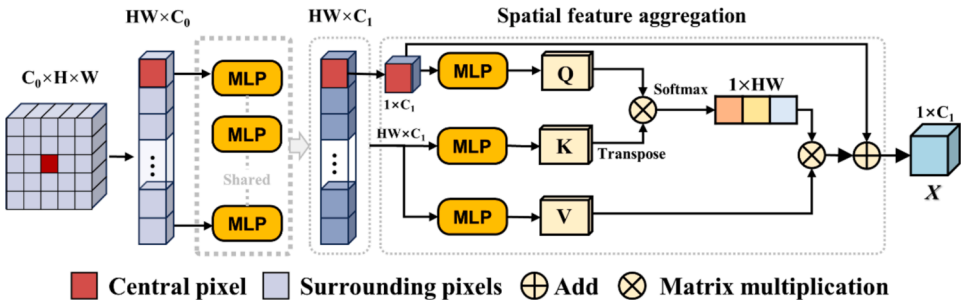


Fig. 3. The architecture of pixel-set aggregation encoder. $H \times W$ represents the patch size, C_0 indicates the number of spectral bands of satellite images, and C_1 is the size of the encoded features derived by the shared-weight MLP layer. Q, K, and V represent query, key, and value vectors generated by three MLP modules, respectively. The comprehensive information on hyperparameters of PSAE is detailed in Part I of Supplementary Material.

these similarity scores and aggregated with the central pixel's features. In general, the PSAE can be viewed as a "spatial attention module" that reduces the influence of potentially irrelevant information from surrounding pixels, thereby learning a more robust spatial representation of the central analyzed pixel.

2.3.2. BiLSTM and temporal attention module

We adopted a two-layer BiLSTM module along with a temporal attention module (Xu et al., 2020) to effectively explore the temporal dependencies of features among different time steps. The architecture of the LSTM cell is illustrated in Fig. 4. Compared to the basic RNN, the LSTM introduces memory cells and gated mechanisms to automatically determine the extent of information retained from previous states and memorized at the current time step. The bidirectional mechanism assists the model in exploring high-level temporal features within the input sequence by processing information in both forward and backward directions. Each encoded hidden feature h_i incorporates information from the preceding (i-1) time steps as well as the subsequent (t-i) time steps (as shown in Fig. 1). This contributes to an enhanced understanding of spectral-temporal variation specific to crop growth. Furthermore, the temporal attention module, which included a fully connected layer with a softmax function, dynamically assigned weights to the hidden features from different time steps for aggregating their information. This enables the model to emphasize important temporal features while suppressing irrelevant features. The computation of attention weights can be expressed by Eqs. (2) and (3):

$$a_t = softmax(FC(h_t, (W, b))) \tag{2}$$

$$h = \sum a_t h_t \tag{3}$$

where h_t represents the hidden features at time step t, and a_t is the corresponding attention weight. W and b are the learnable weight and bias parameters of the fully connected (FC) layer in the attention module, which are shared across different time steps. The softmax function is applied to convert the FC's output into a probability distribution, with these probabilities serving as attention weights that highlight the most relevant time steps in the sequence. h is the final aggregated result obtained by applying these attention weights to the hidden features. Consequently, these aggregated features provide significant advantages in enhancing the model's ability to discern crucial temporal patterns in crops. Based on the highly discriminative spatiotemporal features extracted by BiLSTM and the attention module, the output module predicts the final category information for the central pixel of the input patch. Given that traditional cross-entropy loss may be affected by class imbalance and lead to inferior performance for categories with few samples, we employed focal loss (Lin et al., 2020) as a substitute to adaptively weigh different samples during the training stage. The focal loss can be expressed in Eq. (4):

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{K} y^{i,c}(1-\hat{y}^{i,c})^{\gamma}\log(\hat{y}^{i,c}) \tag{4}$$

where N and K are the number of samples and categories in the dataset, respectively. y and $\hat{y}$ are the ground truth and model prediction, respectively. i and c refer to the index of sample and category, respectively. γ is a tunable parameter for adjusting the model's concentration on "hard examples". A higher γ indicates higher attention of the model on "hard samples". Here, the γ was set as 1. Additionally, to accelerate the model's convergence speed, we adopted the Adam optimizer for the model training (Kingma and Ba, 2015).

2.4. Ablation and comparative experiments

To assess the effectiveness of the developed temporal random masking augmentation and PSAE for Mask-PSTIN in crop classification,

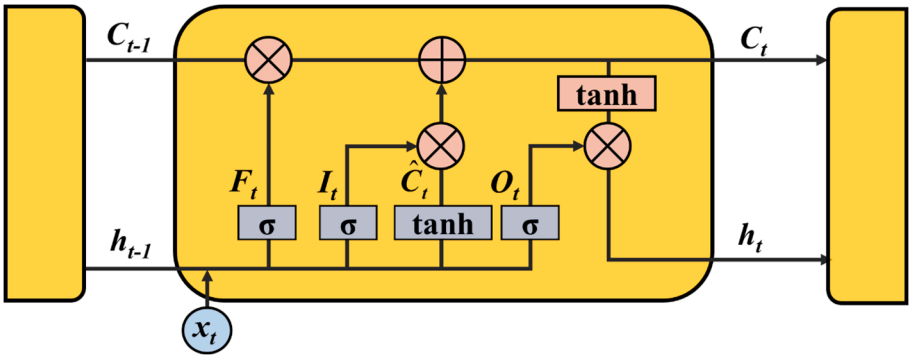


Fig. 4. The structure of LSTM cell. C and $\hat{C}$ refer to the memory cell, and the candidate memory cell, respectively. h and x represent the hidden feature and input feature. F, I, and O are the forget gate, input gate, and output gate, respectively. tanh and σ represent the hyperbolic tangent activation function and sigmoid function, respectively.

we conducted three additional ablation experiments: LSTM, PSTIN, and Mask-LSTM. Notably, the prefix "Mask-" indicates models trained with temporal random masking augmentation. The term "LSTM" refers to the AtBiLSTM model, which excludes the PSAE module from PSTIN and relies solely on the spectral-temporal features of individual pixels. Here, it serves as the baseline classification model. PSTIN improves the LSTM's performance by incorporating additional spatial information, while Mask-LSTM improves it by applying temporal random masking to augment the training data. Mask-PSTIN combines the advantages of both spatial information and temporal random masking, further improving in crop classification.

We also compared the classification results of Mask-PSTIN with two machine learning methods and two deep learning methods. The machine learning methods were random forest (RF) and support vector machine (SVM) (Maulik and Chakraborty, 2017; Mountrakis et al., 2011). RF has two critical parameters of *n_estimators* and *max_features* to determine the number of decision trees and the number of features in each node. According to previous studies (Chen et al., 2023; Melville et al., 2018; Tatsumi et al., 2015), we set *n_estimators* as 500 and *max_features* as the square root of the number of input variables. For SVM, the kernel type is a critical parameter that projects the feature into a high-dimensional space to separate complex non-linear data. Here, we used the radial basis function (RBF) as the kernel of SVM (Foody and Mathur, 2004). Two deep learning methods, including Transformer (Dai, 2019; Xu et al., 2020) and CNN-LSTM, were also employed for comparative experiments. For the Transformer, we configured the hyperparameters as follows: self-attention hidden feature dimension to 512, self-attention heads to 8, self-attention layers to 2, and fully connected layer dimension to 2048. The CNN-LSTM replaces the PSAE module in PSTIN with two 3 × 3 convolutional layers with kernels of 32 and 64 to extract the spatial features of each temporal patch. Besides, the loss function and optimizer settings of these two deep learning methods were aligned with those employed in the Mask-PSTIN to maintain consistency across our experimental framework.

2.5. Accuracy assessment

The confusion matrices were employed for evaluating the performances of different models on both post-harvest and early-season crop type mapping. Three commonly used indicators, i.e., producer's accuracy (PA), user's accuracy (UA), and overall accuracy (OA) were adopted to quantitative measure the omission errors, commission errors, and overall identification errors, respectively. Besides, we calculated the F1-Score by harmonic averaging UA and PA to indicate model performance on each crop type (Zhong et al., 2019). The mean F1-Score (mF1) across different crop types was also computed to evaluate the robustness of our method across various crop types. Moreover, for early-season crop type mapping, we set the F1-Score exceeding 0.8 as a criterion that the target crops are successfully identified (Asam et al., 2022; Weilandt et al., 2023). F1-Score can be calculated as Eq. (5).

$$F1-score = 2 \times \frac{UA \times PA}{UA + PA} \tag{5}$$

3. Study area and data

3.1. Study area

To evaluate the performance of our developed Mask-PSTIN model in post-harvest and early-season crop mapping, we selected three heterogeneous regions with different cropping patterns and agricultural

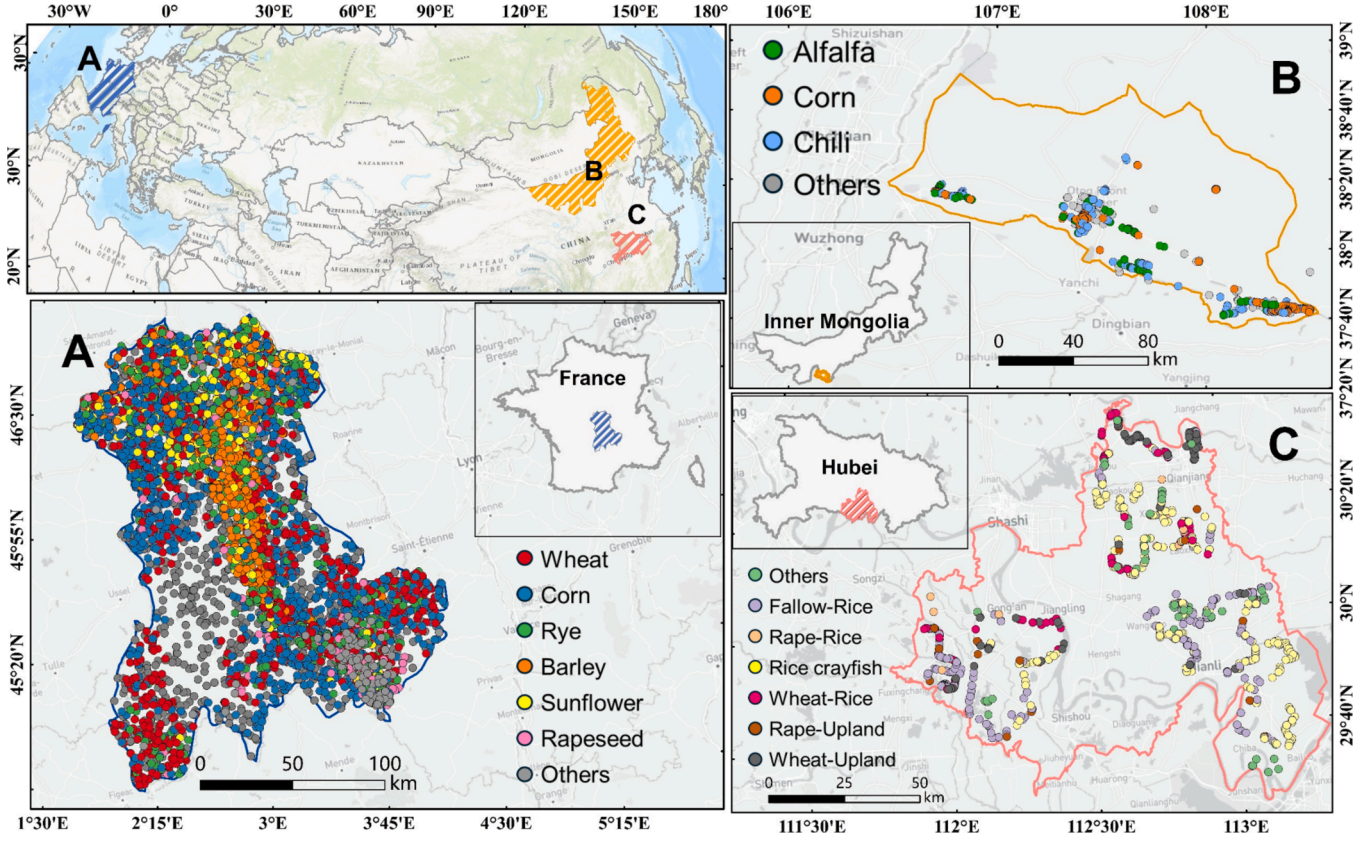


Fig. 5. The geographical location of three study regions and collected samples. Region A is Auvergne situated in central France. Region B corresponds to Otog Front Banner in the southwestern Inner Mongolia Autonomous Region, China. Region C represents five counties, located in Jianghan Plain, in central Hubei Province, China. Notably, the crop rotations presented in region C are as follows: fallow-rice (fallow land-rice rotation), rape-rice (rapeseed-rice rotation), wheat-rice (wheat-rice rotation), rape-upland (rapeseed-upland crop rotation), and wheat-upland (wheat-upland crop rotation).

landscapes (Fig. 5). The phenological calendars of the main crop types across the three study regions are shown in Fig. 6.

Region A (44.6°N – 46.8°N, 2°E – 4.5°E) is Auvergne situated in central France, including four departments: Allier, Puy-de-Dôme, Cantal, and Haute-Loire. It has a temperate marine climate with an average annual temperature and precipitation of 12 °C and 510 mm, respectively. Region A covers a total area of 2.6×10^6 ha, where cropland dominates 74.46 % of the land. The average size of agricultural field parcels is around 2.37 ha. The main crop types in this region include wheat, corn, rye, barley, sunflower, and rapeseed. Winter crops such as wheat, rye, barley, and rapeseed are generally planted in November and harvested in August. Spring crops which include corn and sunflower are usually planted between March and April and harvested between July and October.

Region B (37.62°N – 38.84°N, 106.48°E – 108.54°E) is Otog Front Banner, located in the southwestern Inner Mongolia Autonomous Region, China. It has a temperate continental climate with an average annual temperature of 8.05 °C and an average annual precipitation of 275.83 mm. Rainfall in region B is mainly concentrated from July to September, accounting for about 60 % of the annual precipitation. Region B covers a total area of 1.22×10^6 ha, with agricultural field parcels averaging approximately 1.16 ha in size. The primary crop types are corn, chili, and alfalfa. Specifically, corn and chili share similar phenological dynamics, which are generally planted in May and June and harvested between August and September. Alfalfa is typically planted in March and harvested in September.

Region C (29.43°N – 30.65°N, 111.8°E – 113.32°E) is part of the Jianghan Plain in central Hubei Province, China, including five counties: Qianjiang, Jianli, Gongan, Shishou, and Jiangling. This region covers approximately 1.02×10^6 ha, with cropland constituting around 60 % of the area. Region C experiences a humid continental climate, characterized by an average annual temperature of 17 °C and an average annual precipitation of 1300 mm. The frequent cloud cover and rain pose challenges for accurately identifying crop types, especially in the early season. Moreover, this region is characterized by fragmented agricultural landscapes, with an average field parcel size of approximately 0.51 ha. The major cropping systems include fallow land-rice rotation (fallow-rice), rapeseed-rice rotation (rape-rice), rice crayfish co-culture, wheat-rice rotation (wheat-rice), rapeseed-upland crop rotation (rape-upland), and wheat-upland crop rotation (wheat-upland). Winter crops such as rapeseed and wheat are generally planted in November and harvested in May, while summer crops including mainly rice, are planted between April and May, and harvested in September.

In each study region, we randomly delineated 100 cropland parcels and analyzed their statistical distribution of their *area* and shape index (*shin*) (Cai et al., 2022). As shown in Fig. 7, these three regions, characterized by distinct parcel shapes and sizes, provide a suitable basis for evaluating the generalizability and robustness of Mask-PSTIN model across diverse agricultural landscapes and different crop types.

3.2. Satellite data

The Sentinel-2 Level-2A orthorectified surface reflectance dataset (L2A) in three study regions was downloaded via the Google Earth Engine platform ("COPERNICUS/S2_SR_HARMONIZED"). The L2A images have been processed with radiometric calibration, geometric correction, and atmospheric correction. According to the phenological calendars of crop types in each study area (Fig. 6) and the ground sampling date, the L2A images in the regions A, B, and C were collected from 2020–11-1 to 2021–11-1, 2021–5-1 to 2021–10-27, and 2020–10-1 to 2021–10-1, respectively. Five spectral bands (i.e. blue, green, red, near-infrared, and short-wave infrared 1 bands) were selected to capture the complex spatial details in heterogeneous areas. We resampled those bands to the 10 m spatial resolution and removed cloud-contaminated pixels with a pixel quality score lower than 0.65 based on the cloud score layer for L2A products ("GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED").

Finally, we chose a 20-day composite period to generate the Sentinel-2 image time series (Lin et al., 2022; You and Dong, 2020). Fig. 8 presents the temporal dynamics of NDVI for different crop types derived from time-series satellite images. Due to the variation of crop phenology and impacts of cloudy and rainy weather, the time-series information of crops exhibited complex temporal patterns and considerable noise, introducing uncertainties for accurate crop type mapping.

3.3. Ground truth data

In region A, crop type information for 2021 was obtained from the Registre Parcellaire Graphique (RPG) database (https://geoservices.ign.fr/rpg). The RPG is an open-accessed georeferenced vector dataset derived via field survey, which was released by the Agency for Services and Payment (ASP). The crop type information of RPG database was recorded by farmers (Tommaso et al., 2021). Since RPG data have been widely used as references for crop type mapping in various scientific studies (Lin et al., 2022; You et al., 2023), we also utilized this dataset to construct training and testing datasets. In region A, the RPG database contained 177 categories in 2021, which we consolidated into seven broader categories: wheat, corn (including both silage corn and grain corn), rye, barley, sunflower, rapeseed, and others (including other crop

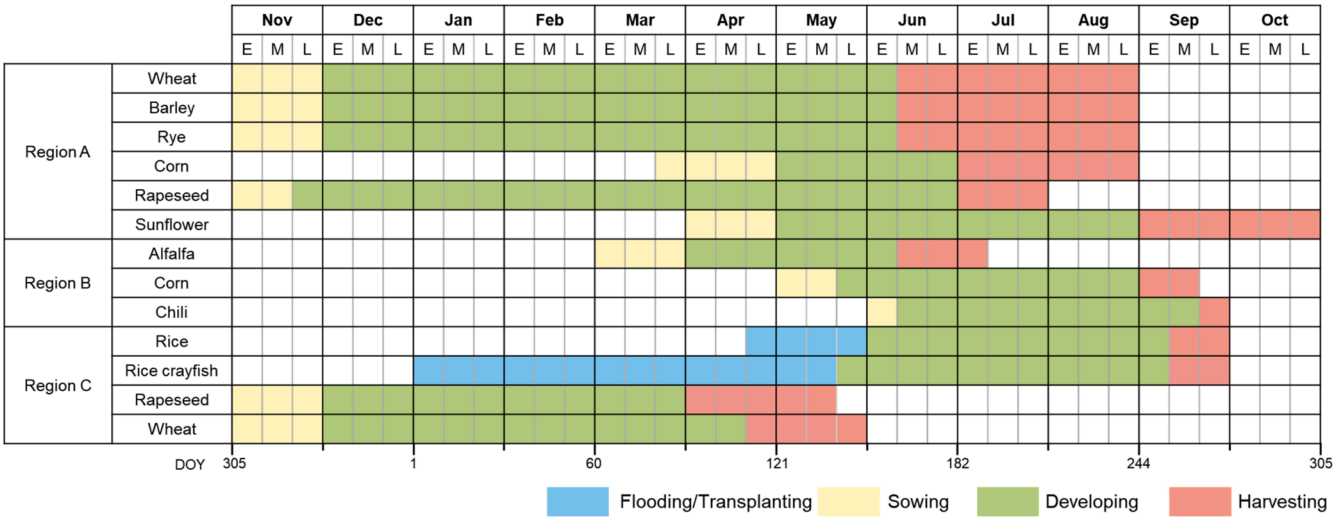


Fig. 6. The phenological calendars of major crop types in three study regions.

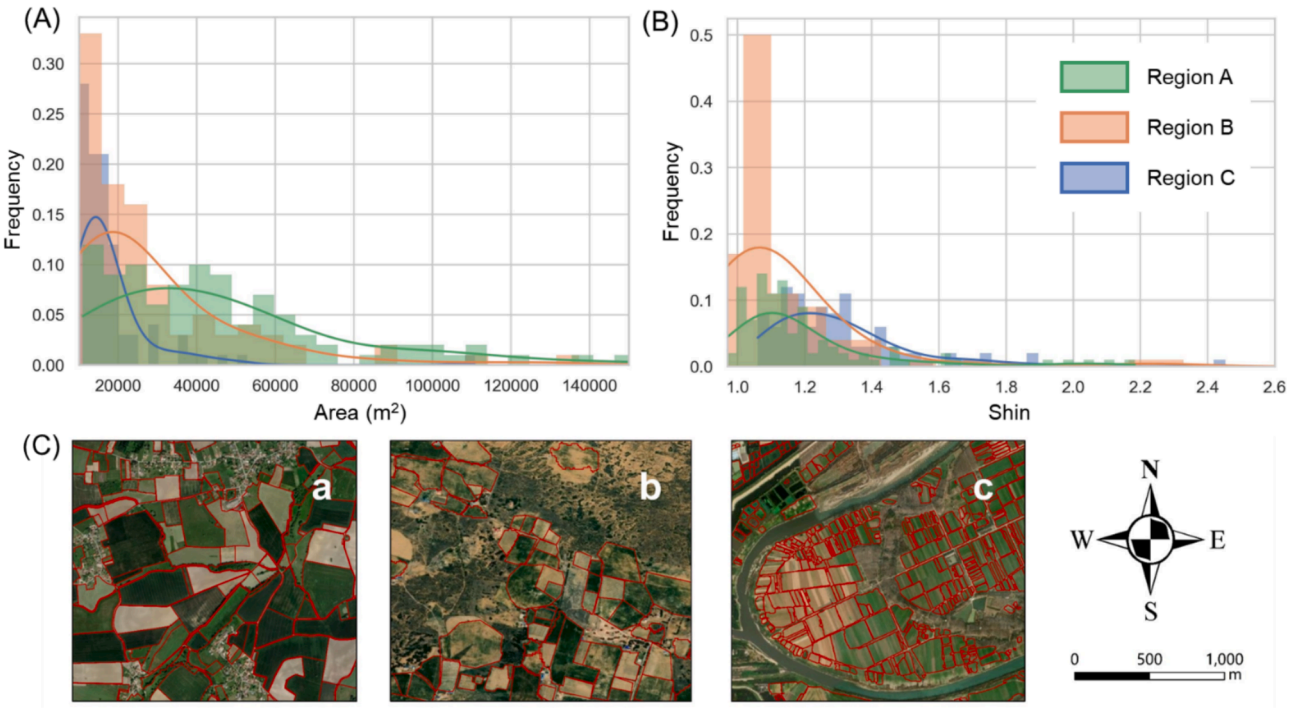


Fig. 7. Statistical distribution of cropland parcels' *Area* (A) and *Shin* (B) in different regions. (C) presents three examples of agricultural field parcels (a, b, c) within the regions A, B, and C. Notably, the *Shin* is calculated by equation of $perimeter/(4 \times \sqrt{area})$. A *Shin* value closer to 1 indicates that the field parcel is more regularly shaped.

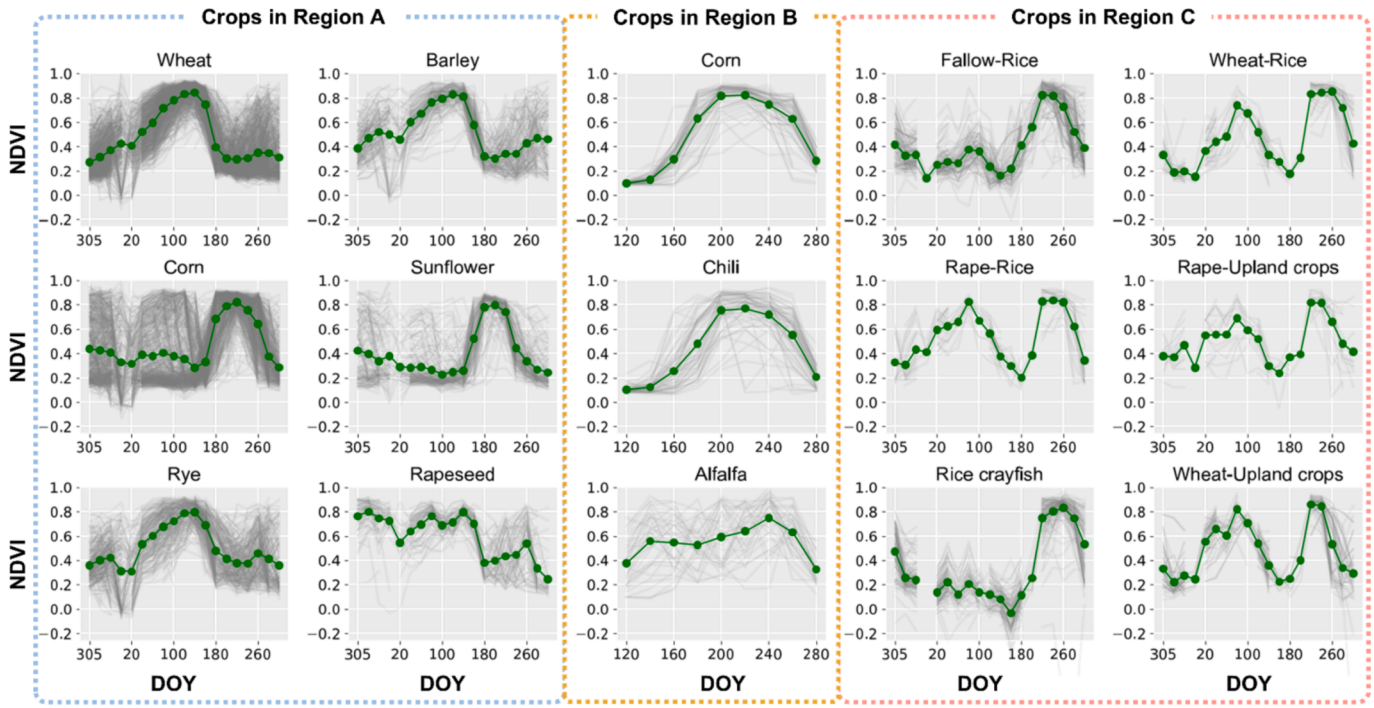


Fig. 8. Temporal dynamic of NDVI for different crop types based on Sentinel-2 image time series. Each gray line represents the time-series NDVI of a field sample, while the solid green lines denote the mean NDVI for all field samples. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and non-crop types), based on the proportion of their planting areas. In total, 7403 samples were collected and split into training and testing datasets in a ratio of 7:3 (Table 1). Among the 5235 training samples, we randomly selected 1919 samples as validation datasets for early stopping during training. The spatial distribution of collected samples is presented in Fig. 5A.

For regions B and C, due to the lack of open-access crop type datasets in China, ground truth information was collected through field surveys in 2021. Region B primarily grows single-season crops, while region C is characterized by complicated rice-cropping rotation systems. Thus, a

Table 1
The number of point samples (i.e., labeled pixels) for the training dataset and testing dataset in regions A-C.

Region	Type	Training	Testing	Total number
Region A	Wheat	1478	614	7403
	Corn	1090	446	
	Rye	535	166	
	Barley	291	116	
	Sunflower	261	95	
	Rapeseed	116	46	
	Others	1464	685	
Region B	Alfalfa	37	26	510
	Chili	66	66	
	Corn	67	75	
	Others	84	89	
Region C	Fallow-Rice	160	105	886
	Rape-Rice	16	25	
	Rice crayfish	140	82	
	Wheat-Rice	37	29	
	Rape-Upland	21	10	
	Wheat-Upland	74	53	
	Others	84	50	

field survey was conducted in June 2021 for region B, and two repetitive field surveys were conducted in April and September 2021 for region C to accurately obtain crop type information. Using handheld GPS devices with a positioning precision error of less than 5 m, we recorded the spatial location and crop type information for each sample. The field surveys followed roads (examples of the survey route can be found in Fig. S1 of Supplementary Material) with samples distributed along roadsides considering the topography, traffic accessibility, and hydrology. The non-crop samples were visually interpreted based on Google Earth. A total of 510 samples were collected in region B and 886 samples in C. Samples from region B were split evenly into training and testing datasets (5:5), while those from region C were split 60 % for training and 40 % for testing (Table 1). Due to the limited number of samples in these two regions, a five-fold cross-validation strategy was employed to tune the model's hyperparameters during the training process.

4. Results

4.1. Ablation experiment of Mask-PSTIN in crop classification

Table 2 presents the OA and mF1 of four models (i.e., LSTM, Mask-LSTM, PSTIN, and Mask-PSTIN) in crop type classification across the three study regions. The Mask-PSTIN model demonstrates remarkable robustness and superior performance, achieving the highest average OA of 87.4 % and mF1 of 0.865. The LSTM model, serving as the baseline, derived the lowest average OA of 80.9 % and mF1 of 0.781. Compared to basic LSTM, the Mask-LSTM model that trained with temporal random masking augmentation effectively reduced the effects of overfitting, resulting in a 3.6 % improvement in average OA and a 0.038 increase in average mF1. The PSTIN model, which aggregated spectral information from neighboring pixels through the PSAE module, also shows similar improvement, with average OA and mF1 increased by about 3 % and 0.037, respectively.

Table 2
Evaluation results of using temporal random masking and pixel-set aggregation encoder for the attention BiLSTM (LSTM) in identifying crop types. The best classification performances are highlighted in bold. Notably, the optimal masking thresholds for each region, determined through parameter analysis in Part III of the Supplementary Material, are 0.40, 0.10, and 0.25, respectively.

		LSTM	Mask-LSTM	PSTIN	Mask-PSTIN
Temporal random masking			✓		✓
Pixel-set aggregation encoder				✓	✓
OA (%)	Region A	80.9 %	83.3 %	81.7 %	**86.2 %**
	Region B	80.1 %	83.2 %	83.2 %	**85.9 %**
	Region C	81.7 %	87.0 %	86.8 %	**90.1 %**
Average OA (%)		80.9 %	84.5 %	83.9 %	**87.4 %**
mF1	Region A	0.767	0.804	0.797	**0.855**
	Region B	0.801	0.838	0.832	**0.863**
	Region C	0.776	0.817	0.826	**0.878**
Average mF1		0.781	0.820	0.818	**0.865**

Fig. 9 and Fig. 10 provide detailed evaluation results and classification maps of all 15 crop types across three study regions. The basic LSTM model shows the worst performance with the lowest F1-Score values (Fig. 9) and undesired "salt and pepper" noise in classification maps (e.g., Fig. 10c1), highlighting the significance of incorporating the temporal random masking and PSAE module. Specifically, incorporating temporal random masking during model training significantly improves its generalization for crop type classification, particularly for crop types with insufficient samples. For instance, in region A, where only 116 training samples were available for rapeseed, Mask-LSTM and Mask-PSTIN significantly increased the F1-score from 0.829 to 0.907, and from 0.867 to 0.933, respectively, compared to LSTM and PSTIN. Adding the PSAE module in PSTIN significantly reduced the "salt and pepper" noise (Fig. 10a2 and c2) in classification maps and demonstrated an improved classification accuracy by leveraging additional spatial context, particularly for distinguishing crops shared very similar phenological characteristics (e.g., barley, rye, and wheat in region A). For example, incorporating spectral-spatial information of adjacent pixels in PSTIN significantly increased the F1-Score of barley by 0.118 compared to LSTM. By integrating temporal random masking augmentation with pixel-set spatial information, the Mask-PSTIN model achieved the highest accuracies for all crop types, indicating its robustness and effectiveness in classification across diverse crop types and agricultural landscapes.

4.2. Comparative experiment results for identifying crop types

Table 3 presents the evaluation results of different classification models across three study regions. The Mask-PSTIN model significantly outperformed the RF, SVM, Transformer, and CNN-LSTM models in identifying various crop types, achieving the highest OA and mF1 of 86.2 % and 0.855 in region A, 85.9 % and 0.863 in region B, and 90.1 % and 0.878 in region C. Among comparative methods, traditional machine learning models (i.e., RF and SVM) performed the worst, primarily due to their limited ability to extract deep abstract features from time-series satellite data. Transformer model demonstrated significant advantages in crop type mapping over RF and SVM in regions B and C, showing notable improvements in both OA and mF1. However, in region A, where crops such as wheat, rye, and barley exhibited very similar phenological features (Fig. 8), Transformer model, which relies solely on spectral-temporal information of individual pixels, struggled to accurately distinguish these crop types. Consequently, it derived inferior performance compared to the CNN-LSTM, which incorporates spectral information of adjacent pixels for crop type classification.

Fig. 11 presents the F1-Score for various crop types, providing a detailed comparison of the performance of five methods across the three study regions. The Mask-PSTIN consistently outperformed RF, SVM, Transformer, and CNN-LSTM, with the highest F1-Score for almost all crop types. Notably, while other classification methods struggled to accurately identify crop types with similar phenological characteristics (such as rye and barley in region A), Mask-PSTIN demonstrated significant improvements by leveraging spatial feature aggregation and temporal random masking technique. Overall, these results highlight the advantages of Mask-PSTIN in classifying diverse crop types with intricate phenology patterns.

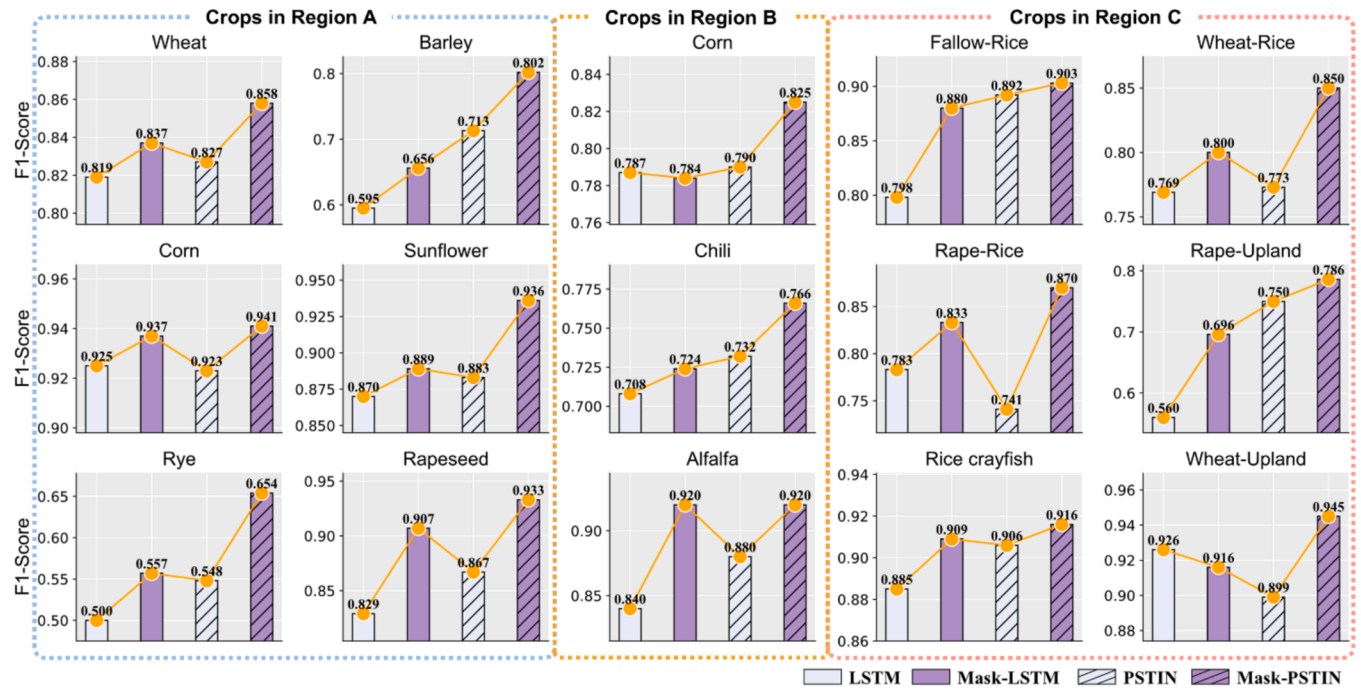


Fig. 9. The F1-Score of LSTM, Mask-LSTM, PSTIN, and Mask-PSTIN for identifying different crop types in three study regions.

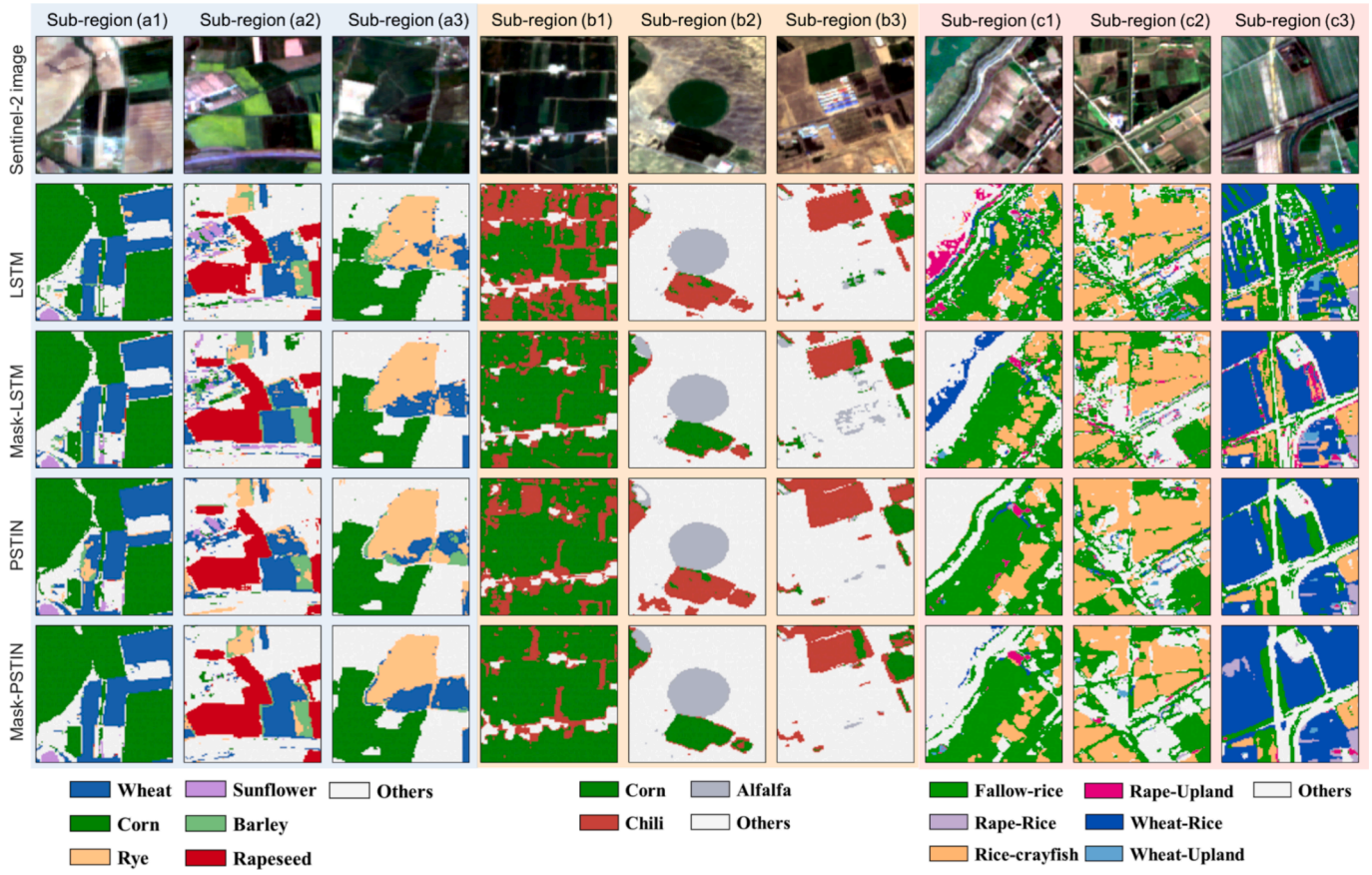


Fig. 10. Crop classification maps derived by LSTM, Mask-LSTM, PSTIN, and Mask-PSTIN in three study regions. (a1-a3), (b1-b3), and (c1-c3) denote sub-regions with areas of 1 × 1 km^2 in region A, region B, and region C, respectively.

Table 3
The overall accuracy (OA) and mean F1-Score (mF1) of different models for crop type classification in three regions. The best classification performances are highlighted in bold.

		RF	SVM	Transformer	CNN-LSTM	Mask-PSTIN
OA (%)	Region A	82.4 %	78.0 %	81.3 %	84.6 %	**86.2 %**
	Region B	82.0 %	79.3 %	85.0 %	84.8 %	**85.9 %**
	Region C	85.9 %	85.1 %	86.8 %	88.7 %	**90.1 %**
mF1	Region A	0.785	0.693	0.763	0.835	**0.855**
	Region B	0.813	0.790	0.844	0.848	**0.863**
	Region C	0.818	0.814	0.839	0.871	**0.878**

4.3. Performance of the Mask-PSTIN in early-season crop mapping for different regions

To assess the performance of Mask-PSTIN in early-season crop mapping, we conducted experiments across ten crop types in three study regions using varying temporal information. Additionally, RF and SVM were employed as reference models due to their widespread use in early-season crop mapping. The F1-Socre variation for early-season crop type identification as the temporal information increases on Mask-PSTIN, RF, and SVM are presented in Fig. 12, where the earliest identifiable times are marked using purple-shaded, yellow-shaded, and blue-shaded, respectively. Overall, all three models exhibited a similar tendency that the accuracy (F1-Score) of crop type identification improves gradually as more temporal information was involved. However, by incorporating spatial information and augmenting training data with temporal random masking technique, Mask-PSTIN demonstrated significant advantages in leveraging limited temporal information and achieved the earliest identifiable time and best accuracies for almost all crops across three regions. For instance, Mask-PSTIN successfully identified wheat, rapeseed, and sunflower in region A during their middle development stage, significantly earlier than two traditional machine learning methods. Furthermore, while Mask-PSTIN, RF, and SVM identified corn in region A on the same day, Mask-PSTIN exhibited a higher F1-Score than those of RF and SVM. In region B, Mask-PSTIN achieved the earliest identifiable time for corn at DOY 200, which was 20 days ahead of RF (i.e., DOY 220). These exceptional results indicate that Mask-PSTIN effectively leveraged limited spectral-spatial–temporal information during the early season, providing a more comprehensive understanding of crop characteristics, and thereby enhancing the model's ability to achieve significantly earlier crop type identification.

5. Discussion

5.1. Explanation of the Mask-PSTIN advantages in crop type mapping

To further understand the efficacy of Mask-PSTIN in classifying different crops, we introduced gradient propagation to assess the importance of input time-series features in deep learning models (Tang et al., 2024; Xu et al., 2021). Since the deep neural networks are fully differentiable functions designed to model the correlations between input time-series features $[X_1, X_2, \ldots X_t]$ and the target crop type y, the gradient can be backpropagated to each element X_i within the input feature sequence. The derivative values signify the influence of each input feature on the model's prediction. Positive (negative) values indicate a positive (negative) correlation between changes in input features and the model's predicted scores, while a zero derivate suggests that the changes in input have no effects on the prediction. Therefore, the absolute gradient values reflect the importance of input features for the model, thereby indicating the model's capability to leverage and explore crucial input features.

Based on the gradient backpropagation, we first computed the absolute gradient values of each input element. Subsequently, we normalized these gradient values to analyze feature importance for wheat and barley in region A, corn in region B, as well as rape-rice and fallow-rice in region C under Mask-PSTIN, PSTIN, Mask-LSTM, and LSTM, as presented in Fig. 13. For the models that used image patches (i.e., Mask-PSTIN and PSTIN), we averaged the gradients across all pixels within a patch to determine the gradient values of each spectral band at each time step. In general, models trained without temporal random masking (i.e., PSTIN and LSTM) were easily overfitted to specific temporal patterns or noise, leading to biased crop representation and reduced classification accuracy. For instance, in region C for fallow-rice, LSTM showed strong dependency on features during DOY 365 to 160 (i.

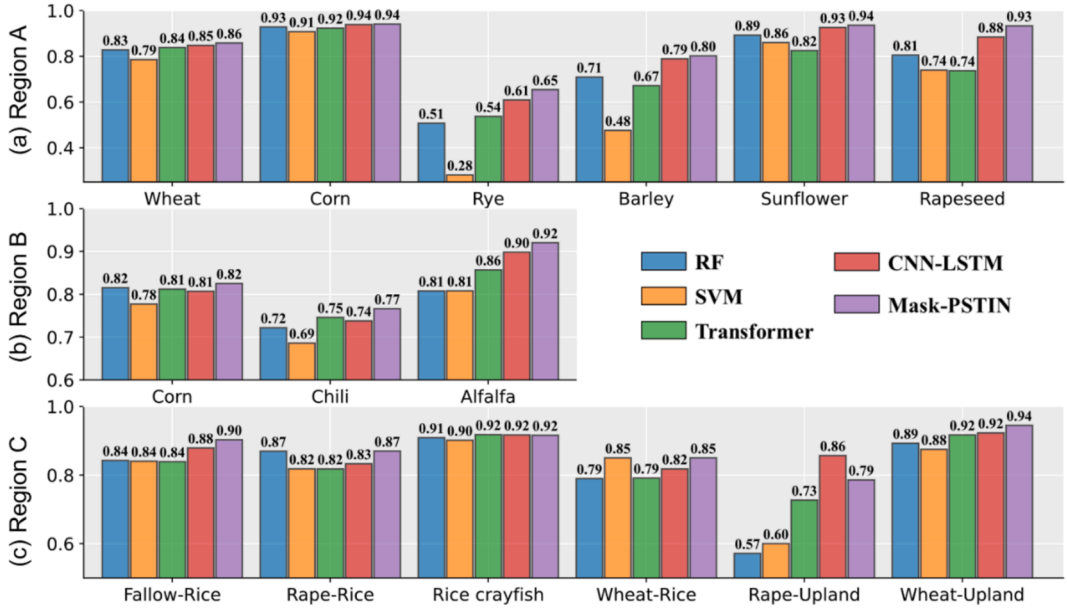


Fig. 11. The F1-Scores of RF, SVM, Transformer, CNN-LSTM, and Mask-PSTIN in identifying different crop types across the three study regions.

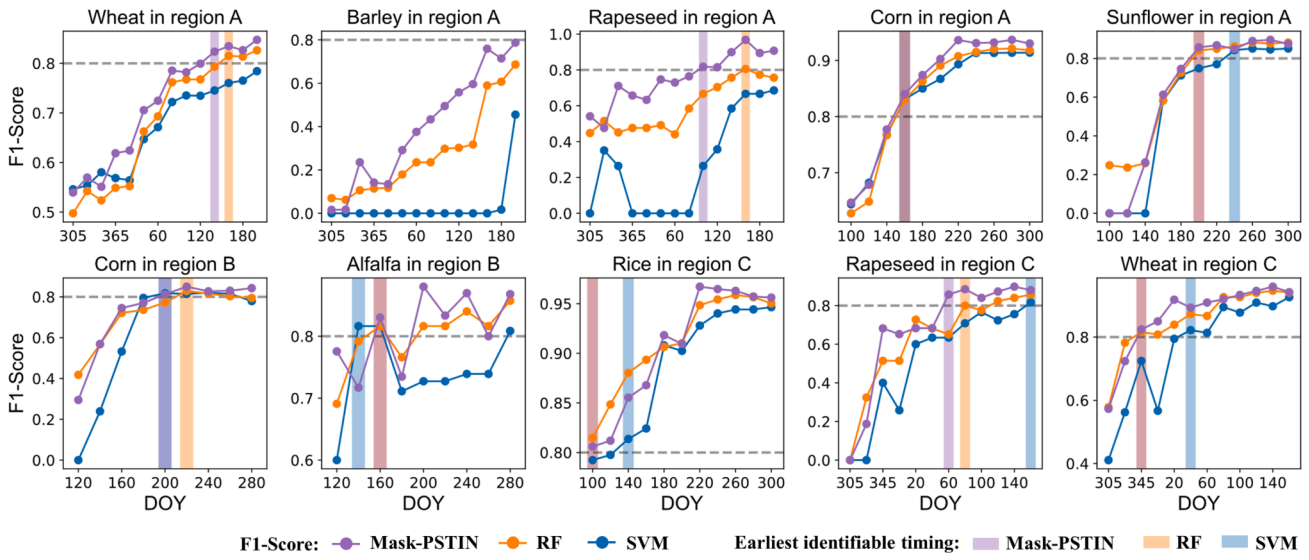


Fig. 12. The F1-Score variation for early-season crop type identification as the temporal information increases using Mask-PSTIN, RF, and SVM models in three regions. Region A included wheat, barley, rapeseed, corn, and sunflower. Region B included corn and alfalfa. Region C included rice, rapeseed, and wheat. The shaded regions of various colors represent the earliest identifiable time (i.e., F1-Score $>$ 0.8) of different models for the target crop types.

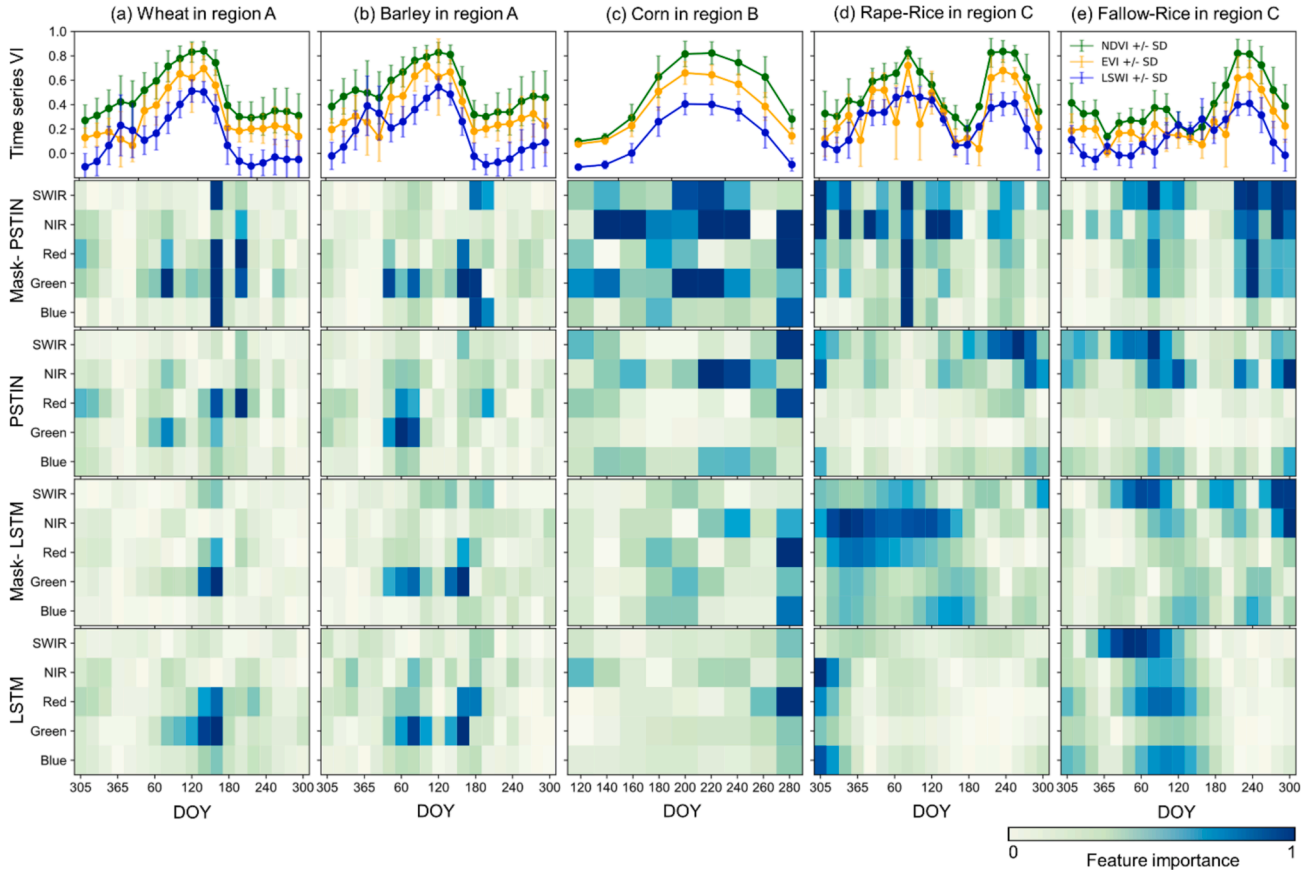


Fig. 13. Feature importance analysis for identifying crop types across Mask-PSTIN, PSTIN, Mask-LSTM, and LSTM models. The feature importance maps were generated by averaging the importance of input features across all testing samples of the corresponding crop types.

e., the fallow stage), while overlooking features during the rice growing stage. Similarly, PSTIN in region C for rape-rice, relied heavily on features during DOY 220 to 380 (i.e., the rice growing stage), neglecting features during the rapeseed growing stage. In contrast, Mask-PSTIN and Mask-LSTM exhibited broader attention across different time steps, and their derived feature importance showed significantly higher consistency with the crops' phenological dynamics. These results demonstrate that employing temporal random masking for data augmentation during training effectively assists the models in learning more robust and invariant features, thereby improving their performance and

adaptability to incomplete or noisy time-series data.

Furthermore, compared to Mask-LSTM (or LSTM) which only uses individual pixel spectral-temporal features, Mask-PSTIN (or PSTIN) is more advantageous in capturing critical phenological features of crops by aggregating spectral information from surrounding pixels, particularly valuable for early-season crop identification. For example, in region C for rape-rice, Mask-LSTM focused more on features during the harvesting stage, whereas Mask-PSTIN extended its attention to the entire growing stages (i.e., DOY 140 to 240). Additionally, the activated time steps for Mask-PSTIN appeared to be sparser, with adjacent time steps exhibiting significantly different importance, such as DOY 40 to 60 for rape-rice in region C. This sparsity can be attributed to the incorporation of patch-level spatial information, which improved the model's understanding of crops and reduced the need for dense time-series data, underscoring its ability to capture key phenological characteristics even with incomplete time-series data.

5.2. Implications and prospect

Our experiments demonstrated that Mask-PSTIN shows significant advantages in effectively extracting intricate spatiotemporal features of different crop types and achieving robust classification results across diverse agricultural landscapes. However, several improvements could be considered in future studies.

Firstly, the masking threshold parameter τ_m plays a crucial role in controlling the extent of data augmentation. Higher τ_m values indicate that more temporal information was masked during training, while lower τ_m values mean less masking. Our parameter analysis showed that models trained with non-zero τ_m consistently outperformed the baseline model without temporal random masking ($\tau_m = 0$) in crop type mapping (see Part III in Supplementary Material). This underscores the effectiveness of temporal random masking augmentation in improving the model's generalizability and robustness for crop type mapping. However, careful tuning of the τ_m remains crucial to make a balance between introducing sufficient data variability to prevent overfitting and retaining enough temporal information for accurate classification. Crops with longer growing seasons typically benefit from a higher τ_m value, which allows for greater data variability while retaining essential temporal information. In contrast, crops with shorter or more complex phenological patterns may require a lower τ_m value to ensure that critical phenological details are preserved while still achieving adequate data augmentation. Generally, a τ_m value between 0.1 and 0.5 is recommended to ensure the proper augmentation while preserving essential temporal information. The τ_m can be further refined and adjusted according to the specific crop types.

Secondly, deep learning models, including Mask-PSTIN, have a far greater number of trainable parameters than traditional machine learning models. Training such models with limited samples can lead to overfitting and performance degradation. Mask-PSTIN derived relatively low accuracy when using very limited training samples (332 samples for 7 crop types in region A) (see Part IV in Supplementary Material). However, by integrating temporal random masking augmentation with pixel-set spatial information, Mask-PSTIN successfully learned more comprehensive and generalized features, consistently outperforming the traditional RF model across various training sample sizes. Notably, Mask-PSTIN achieved comparable accuracies with RF while using significantly fewer training samples (1328 samples:2656 samples). This indicates Mask-PSTIN's potential for large-scale applications, even when sample availability is constrained. Moreover, the rapid development of satellite sensors has provided richer spectral information that can capture the physiological differences among crop types more effectively. Mask-PSTIN demonstrated significant improvements in classification accuracy when incorporating the SWIR1 band alongside the basic spectral bands (red, green, blue, and near-infrared bands) (Part V in Supplementary Material). Remarkably, even with fewer spectral inputs, Mask-PSTIN outperformed RF, which utilized more extensive spectral information. This underscores Mask-PSTIN's superior ability to extract and leverage spatial-temporal features from satellite image time series. Future studies could further enhance the model by incorporating richer spectral information, such as red-edge bands, to handle more complex cropping patterns.

Finally, due to the spatiotemporal variation in crop phenology and the challenges associated with timely crop sample collection, crop type mapping over different time periods and regions remains to be explored, particularly for early-season crop classification. Our spatiotemporal transferability analysis in region A revealed that Mask-PSTIN demonstrated good spatial transferability across almost all crop types, effectively classifying crops when trained in one sub-region and applied to another (Part VI in Supplementary Material). In terms of temporal transferability, the model performed well for crops with distinct phenological features, such as corn, sunflower, and rapeseed. However, it struggled with crops like wheat, rye, and barley, which share very similar phenological traits. This reduced performance, when trained on data from different years, is likely due to year-to-year phenological shifts, making it difficult for the model to distinguish subtle differences in spatiotemporal features among these crop types. Future studies could focus on improving temporal transferability by considering these phenological shifts (Kluger et al., 2021; Pham et al., 2024a).

6. Conclusion

This study developed the Mask-PSTIN that comprises temporal random masking technique and a novel PSTIN model for mapping both post-harvest and early-season crop types. Specifically, the temporal random masking technique improves variability of limited training data by selectively masking observations at certain time steps in the multi-temporal satellite images during the training, enforcing the model to learn more robust and invariant features of crop types. PSTIN combines a novel pixel-set aggregation encoder with an AtBiLSTM module to capture the comprehensive spatiotemporal patterns of time-series satellite image patches. To test the effectiveness of the proposed Mask-PSTIN, we conducted experiments in the Auvergne region of France, and Otog Front Banner in Inner Mongolia Autonomous Region, and five counties in Hubei province of China. The results demonstrated that the combination of temporal random masking augmentation and PSTIN model significantly improved crop identification accuracy compared to basic bidirectional LSTM models, with average OA and mF1 increased by about 6.5 % and 0.084 across three regions, respectively. Additionally, Mask-PSTIN significantly outperformed two traditional machine learning models (RF and SVM) and two deep learning models (Transformer and CNN-LSTM) in crop classification, achieving the highest OA and mF1 of 86.5 % and 0.855 in region A, 85.9 % and 0.863 in region B, and 90.1 % and 0.878 in region C, respectively. Furthermore, for ten crop types across three study regions, Mask-PSTIN achieved accurate crop type mapping in the early season (i.e., before or during the developing stages of crops), and advanced the earliest identifiable time before harvesting compared to RF and SVM. The feature importance derived by gradient backpropagation indicated that Mask-PSTIN was prominent in achieving broader attention for the entire time series features and generating much more generalizable features, which is critical to improve the accuracy of both post-harvest and early-season crop type classification. Overall, the developed Mask-PSTIN model offers a valuable tool for decision-makers seeking accurate and timely spatial information on crop types, holding great potential for supporting crop management activities and growth monitoring.

CRediT authorship contribution statement

Xinyu Zhang: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Zhiwen Cai:** Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Qiong Hu:** Writing – review & editing, Supervision, Methodology,

Conceptualization. **Jingya Yang:** Validation, Methodology. **Haodong Wei:** Validation, Data curation. **Liangzhi You:** Writing – review & editing, Supervision, Conceptualization. **Baodong Xu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (2023YFD1900101) and National Natural Science Foundation of China (42271360, 42271399, 42411540181).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.isprsjprs.2024.10.013.

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